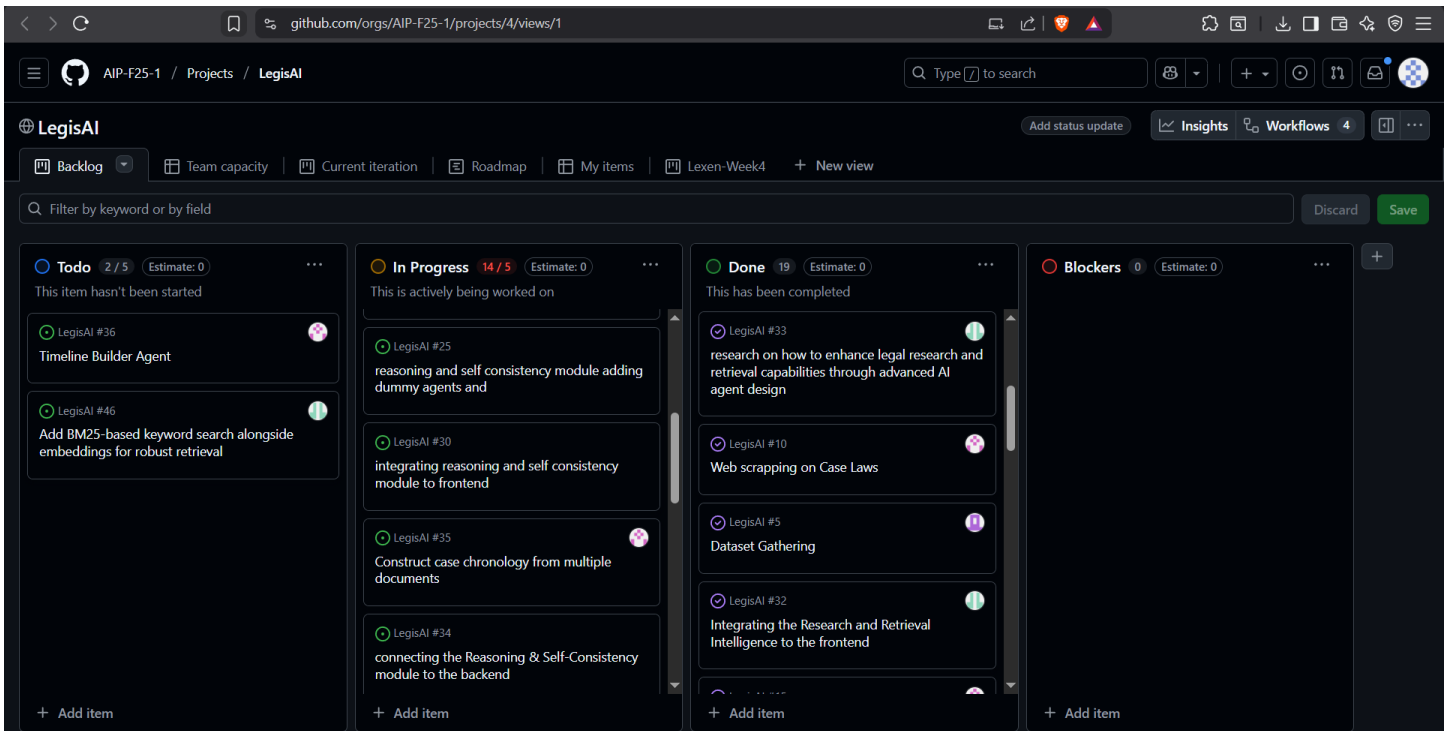
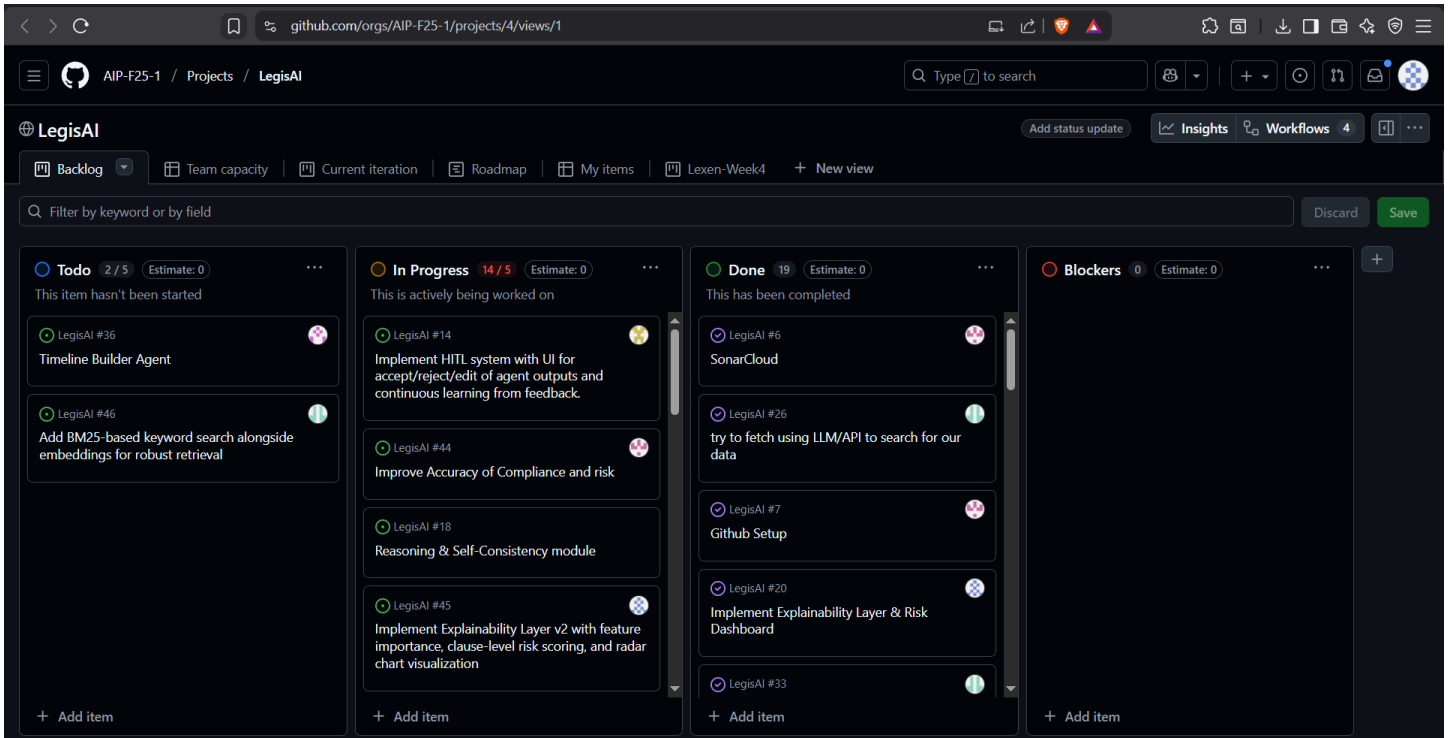


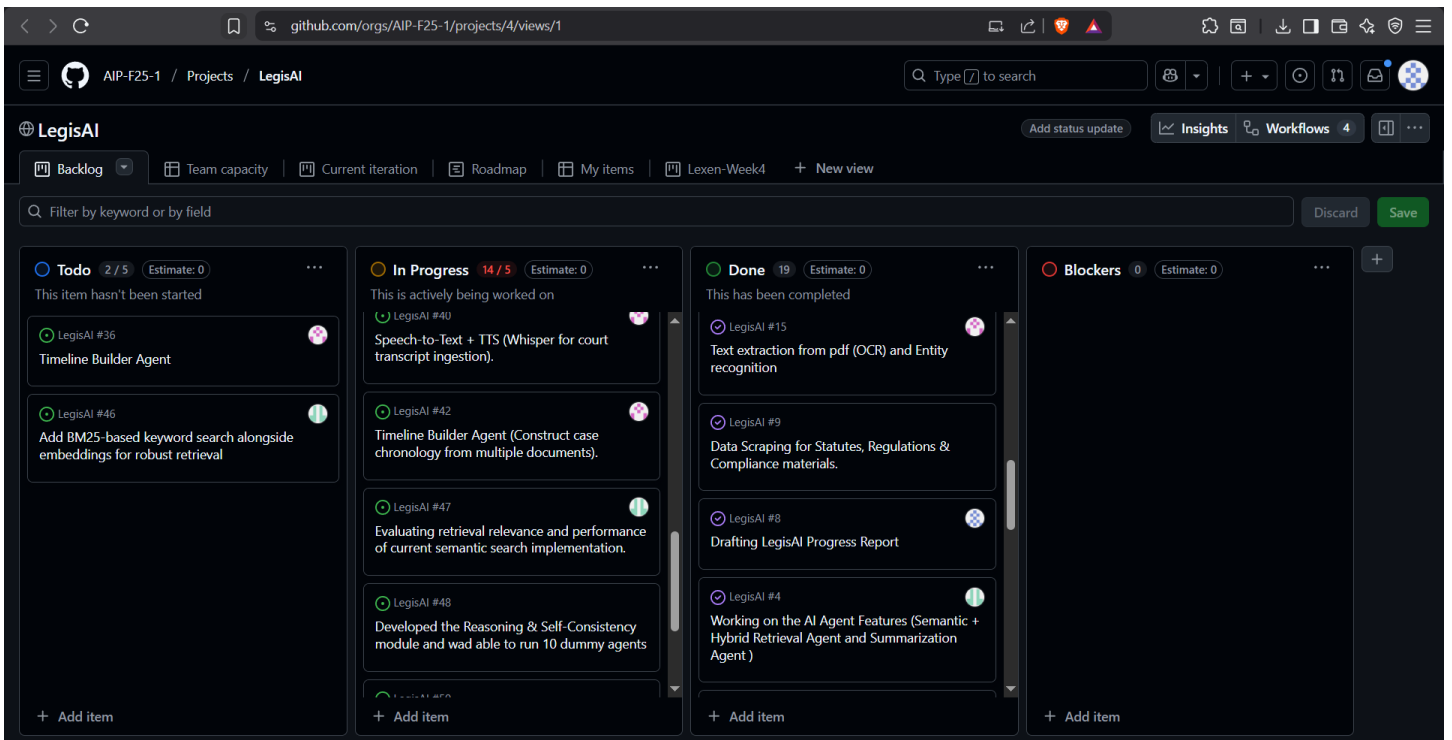
PROGRESS REPORT #9

AICS3 – LegisAI

Project Board Link: [Backlog-LegisAI](#)

Project Board Screenshot(s):

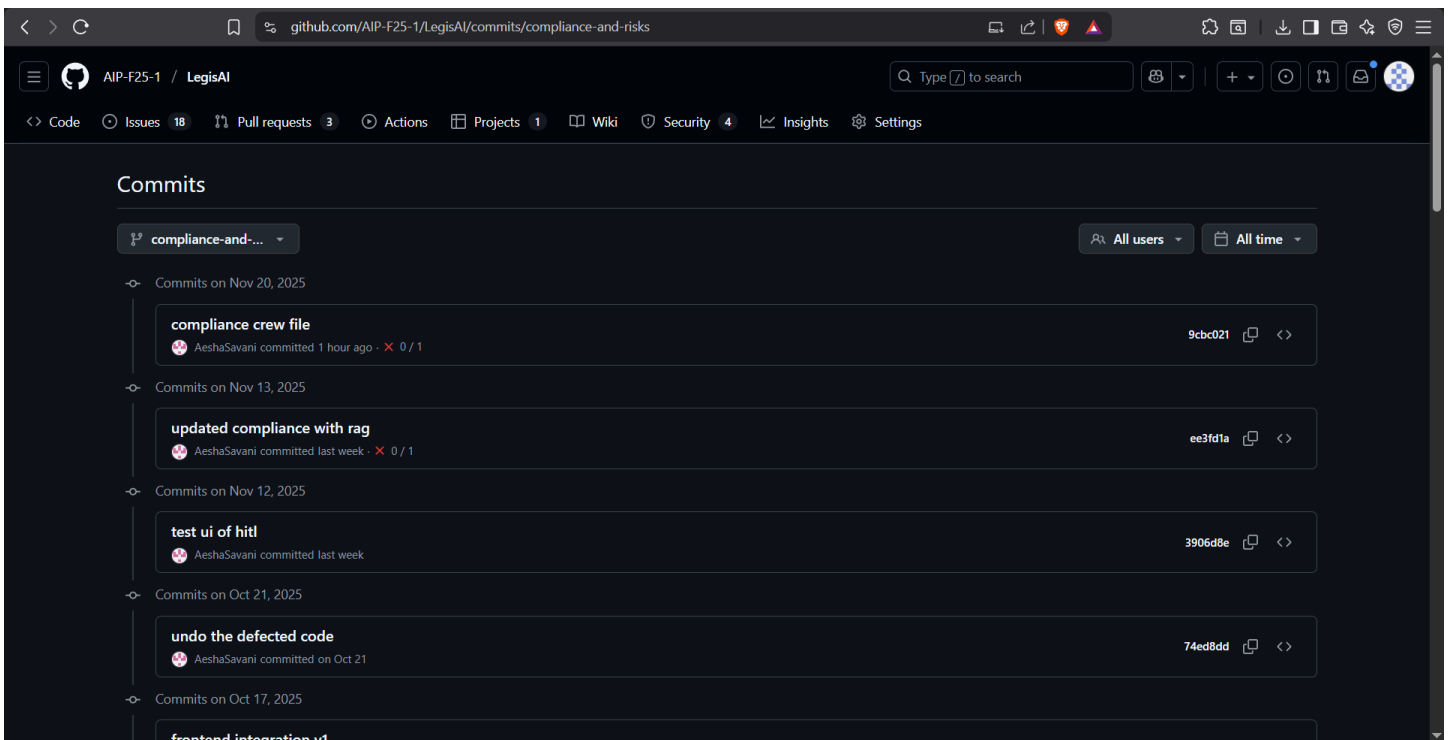




GitHub Repo Link: [AIP-F25-1/LegisAI](https://github.com/AIP-F25-1/LegisAI)

GitHub Commits and dashboard tickets Screenshot(s) for Individual Team Member:

Aesha Mavani



Nithish Janagam

Untitled

github.com

<> Code Issues 18 Pull requests 3 Actions Projects 1 Wiki Security 3 Insights Settings

Commits

nithish

All usersAll time

Commits on Nov 20, 2025

feature update

GITHUB43769 committed 3 minutes ago

4e18cf3

Commits on Oct 30, 2025

frontend features

GITHUB43769 committed 3 weeks ago

4af4b78

Commits on Oct 5, 2025

US_caselaws_scrape

GITHUB43769 committed on Oct 5

77ceaed

Commits on Sep 26, 2025

canadian dataset

GITHUB43769 committed on Sep 26

640ce6b

Commits on Sep 25, 2025

folder skeleton test files added

AeshaSavani committed on Sep 25

4d2a11e

folder skeleton

AeshaSavani committed on Sep 25

841d713

Commits on Sep 16, 2025

Arjun Kumar

github.com/AIP-F25-1/LegisAI/commits/dev

AIP-F25-1 / LegisAI

Type to search

<> Code Issues 18 Pull requests 3 Actions Projects 1 Wiki Security 4 Insights Settings

Commits

dev

All usersAll time

Commits on Nov 20, 2025

Add Explainability Layer v3 with risk labels, CSV export, and aggregated feature importance

Arjun500238195 authored 1 minute ago · 0 / 1

Verified4a1fec

Merge pull request #56 from AIP-F25-1/implement-human-in-the-loop

nidhichakrani authored 2 hours ago · 0 / 1

Verifiede10981b

Commits on Nov 19, 2025

Merge pull request #55 from AIP-F25-1/jeffin_explainability_final

jeffn11 authored yesterday · 0 / 1

Verified585cb7e

Added explainability work + Week 11 deliverables

jeffinJohn1 committed yesterday · 1 / 1

e83002f

Commits on Nov 13, 2025

Merge pull request #53 from AIP-F25-1/Resti_tech_code

Backs authored last week · 0 / 1

Verified0fb28c

Updated the Hybrid Search pipeline, Knowledge Graph Insights, Precedent Reasoning

c954ff5

Jeffin John Abraham

AIP-F25-1 / LegisAI

Type to search

<> Code

Issues 18

Pull requests 3

Actions

Projects 1

Wiki

Security 4

Insights

Settings

Commits

jeffin_explainabil...

All usersAll time

Commits on Nov 19, 2025

Added explainability work + Week 11 deliverables

JeffinJohn1 committed yesterday · ✓ 1 / 1

e83002f

Commits on Nov 13, 2025

Merge pull request #53 from AIP-F25-1/Resti_tech_code

Verified0f8c28c

Updated the Hybrid Search pipeline, Knowledge Graph Insights, Precedent Reasoning

Bacos committed last week · ✓ 1 / 1

c954ff5

Merge pull request #52 from AIP-F25-1/Sumith_Multi_Modal

Verified9f06f55

adding Documentation.py file

Sumith7577 authored and Sumith7577 committed last week · ✓ 1 / 1

528d432

removing files

Sumith7577 authored and Sumith7577 committed last week

305092f

Commits on Nov 12, 2025

Nidhi Chakrani

AIP-F25-1 / LegisAI

Type to search

<> Code

Issues 18

Pull requests 3

Actions

Projects 1

Wiki

Security 4

Insights

Settings

Commits

implement-huma...

All usersAll time

Commits on Nov 13, 2025

create dummy hitl agent for chatbot

nidhichakrani committed last week · ✗ 0 / 1

d5819fe

Commits on Oct 14, 2025

background connection and install dependencies

nidhichakrani committed on Oct 14

740b315

Commits on Oct 10, 2025

setup API key and install dependencies

nidhichakrani committed on Oct 10

d0a9da8

Commits on Oct 7, 2025

Merge pull request #23 from AIP-F25-1/implement-human-in-the-loop

Verified0efe4e3

pkg install

nidhichakrani committed on Oct 7

125eb41

Resti Angelo Cenia

github.com/AIP-F25-1/LegisAI/commits/Resti_tech_code

AIP-F25-1 / LegisAI

Q Type to search

+

<> Code

Issues 18

Pull requests 3

Actions

Projects 1

Wiki

Security 4

Insights

Settings

Commits

Resti_tech_code

All usersAll time

Commits on Nov 13, 2025

Updated the Hybrid Search pipeline, Knowledge Graph Insights, Precedent Reasoning

Backs committed last week - ✓ 1 / 1

c954ff5

<>

Commits on Oct 9, 2025

Updated Research and Retrieval Intelligence

Backs committed on Oct 9

731592f

<>

Commits on Oct 2, 2025

added the knowledge_graph, precedent_agent, statue_entity

Backs committed on Oct 2

c8b1f66

<>

Commits on Sep 26, 2025

added Retrieval_agent

Backs committed on Sep 26

a291319

<>

added Retrieval_agent code

Backs committed on Sep 26

880c047

<>

Sumith Padma

github.com/AIP-F25-1/LegisAI/commits/Sumith_Multi_Modal

AIP-F25-1 / LegisAI

Q Type to search

+

<> Code

Issues 18

Pull requests 3

Actions

Projects 1

Wiki

Security 4

Insights

Settings

Commits

Sumith_Multi_M...

All usersAll time

Commits on Nov 13, 2025

adding Documentation.py file

Sumith7577 authored and Sumith7577 committed last week - ✓ 1 / 1

528d432

<>

removing files

Sumith7577 authored and Sumith7577 committed last week

305092f

<>

Commits on Nov 6, 2025

Speech to text

Sumith7577 authored 2 weeks ago

Verified220f447

<>

Commits on Oct 2, 2025

Moved document file inside agents folder

Sumith7577 committed on Oct 2

83283d6

<>

Merge pull request #17 from AIP-F25-1/Sumith_Multi_Modal

Sumith7577 authored on Oct 2

Verifiede3f6456

<>

frontend skeleton

AeshaSavani committed on Oct 2

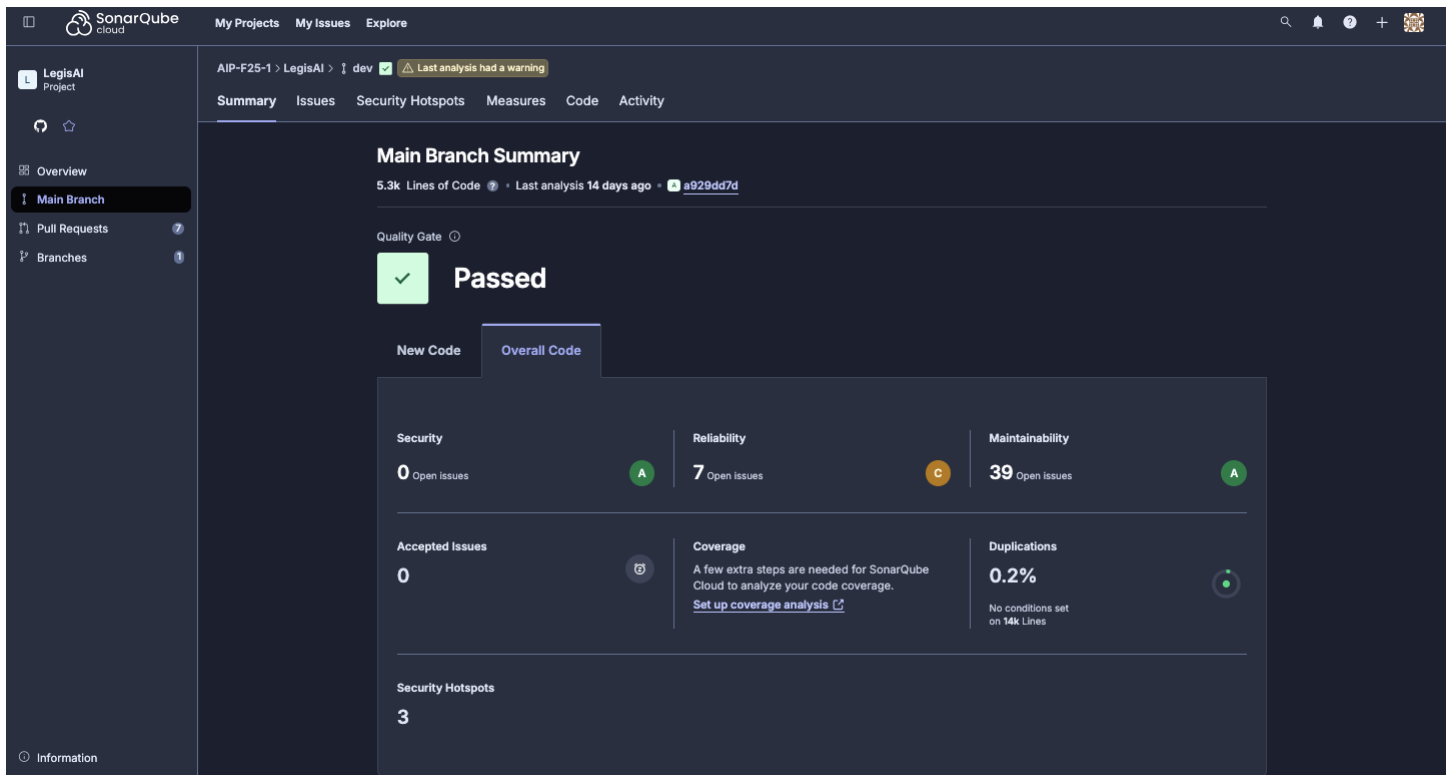
5f5b543

<>

Features developed so far:

- Research and retrieval intelligence agent
- Drafting and contract intelligence agent
- Compliance and risk intelligence agent
- Explainability Layer implementation

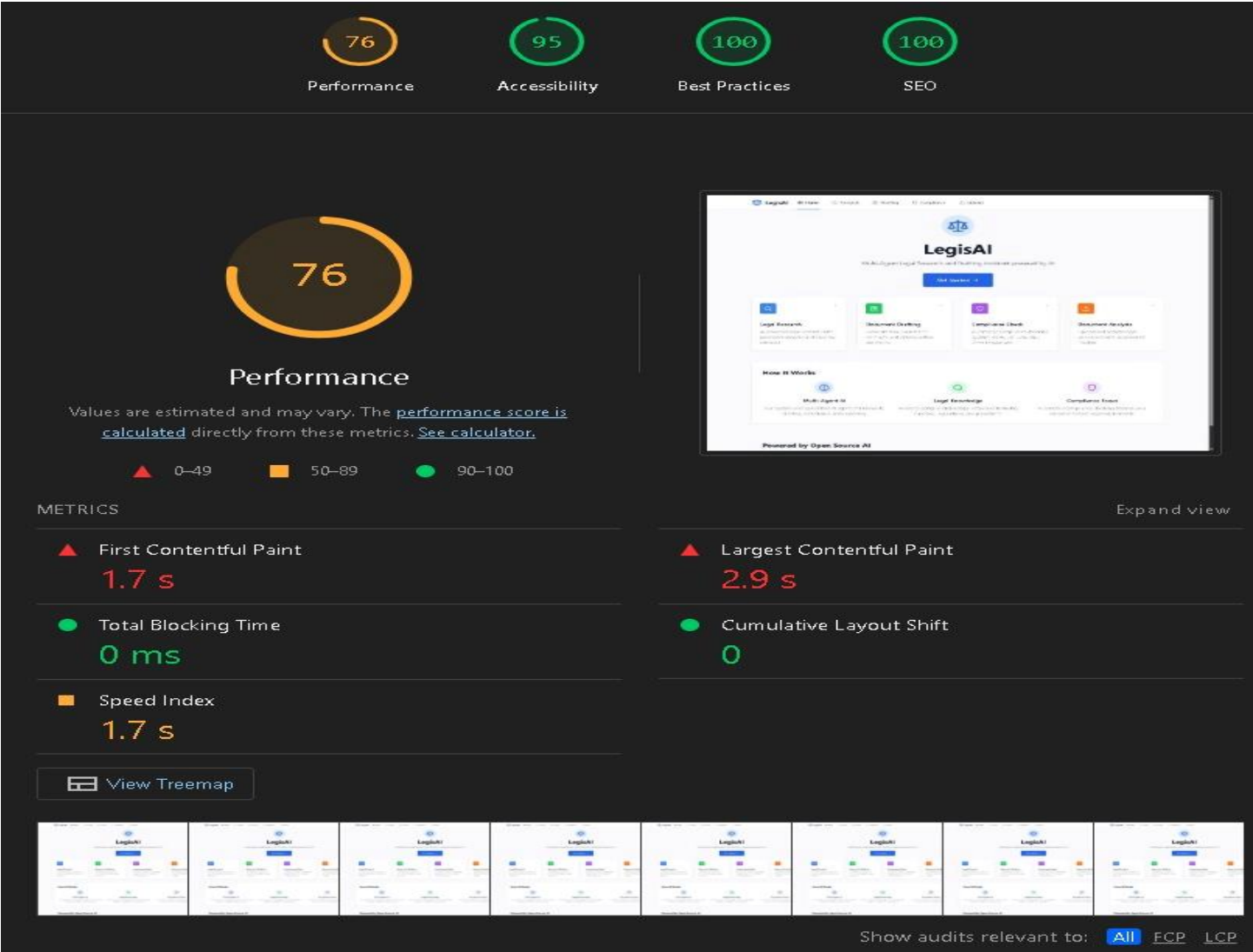
SonarQube Screenshot(s):



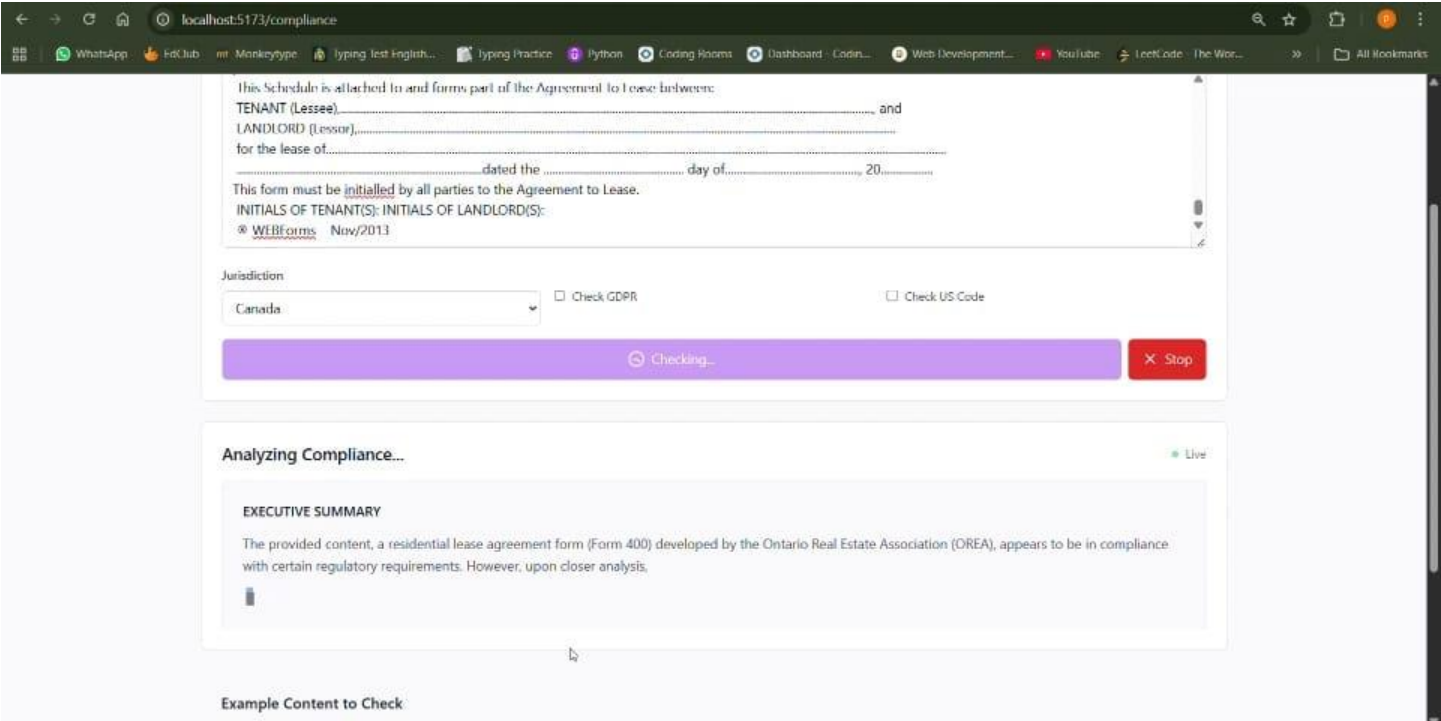
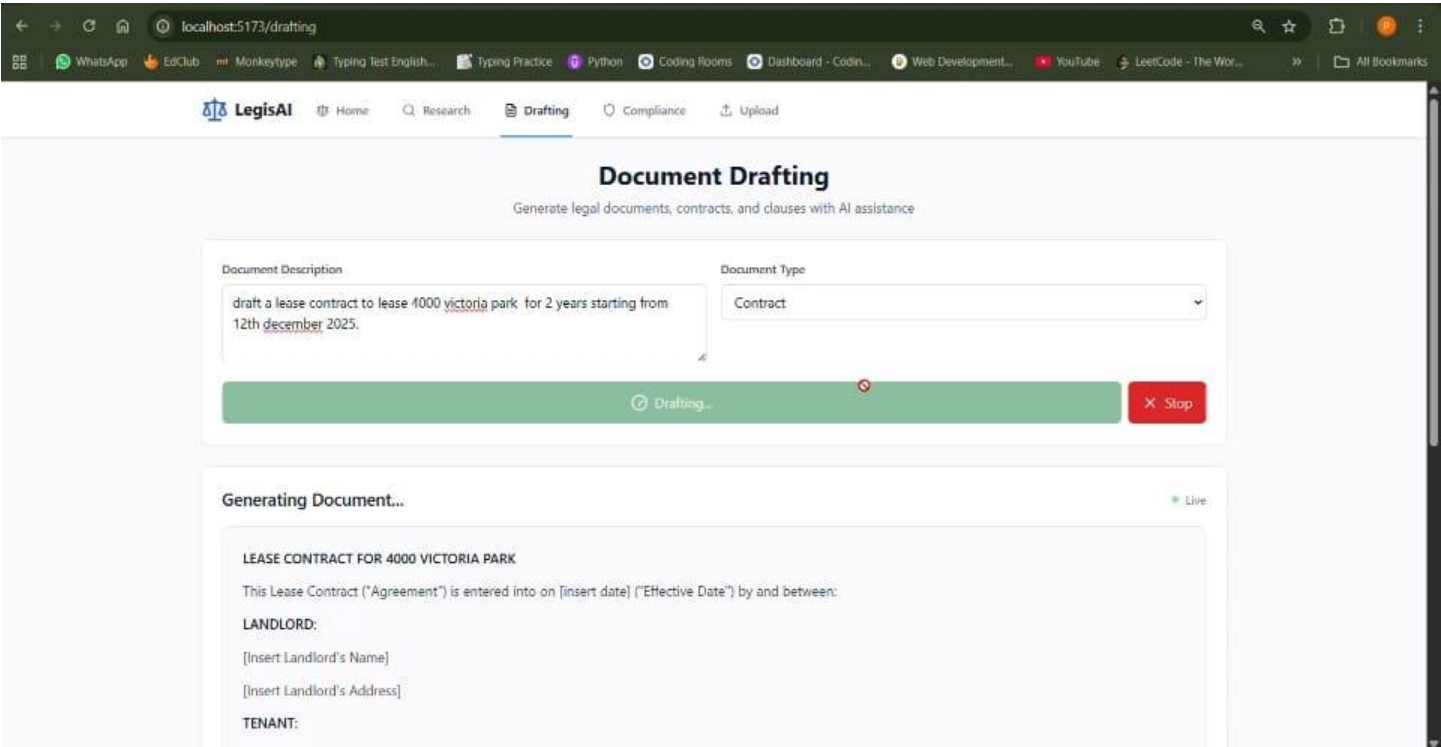
Team Member	Task	Effort Hours	Cumulative Hours	Status
Aesha Mavani	Developed a 30-second marketing video for LegisAI and performed extensive SonarQube compliance fixes for the Compliance & Risk module	42	100	In Progress
Nithish Janagam	Expanded existing features by adding complementary sub-features to improve overall system functionality.	50	100	In progress
Arjun Kumar	Develop Explainability Layer v3 with risk labeling, aggregated feature importance, and CSV reporting	40	80	In Progress
Jeffin John Abraham	Designed the foundational structure of the chatbot module and developed the initial flow for processing user queries	30	90	In Progress
Nidhi Chakrani	Built and tested Human-in-the-Loop (HITL) dummy agents to study human feedback interaction and began integrating the HITL mechanism into a live chatbot	30	90	In progress
Resti Angelo Cenia	Retrieval relevance for evaluating the performance of current semantic search	55	95	In Progress

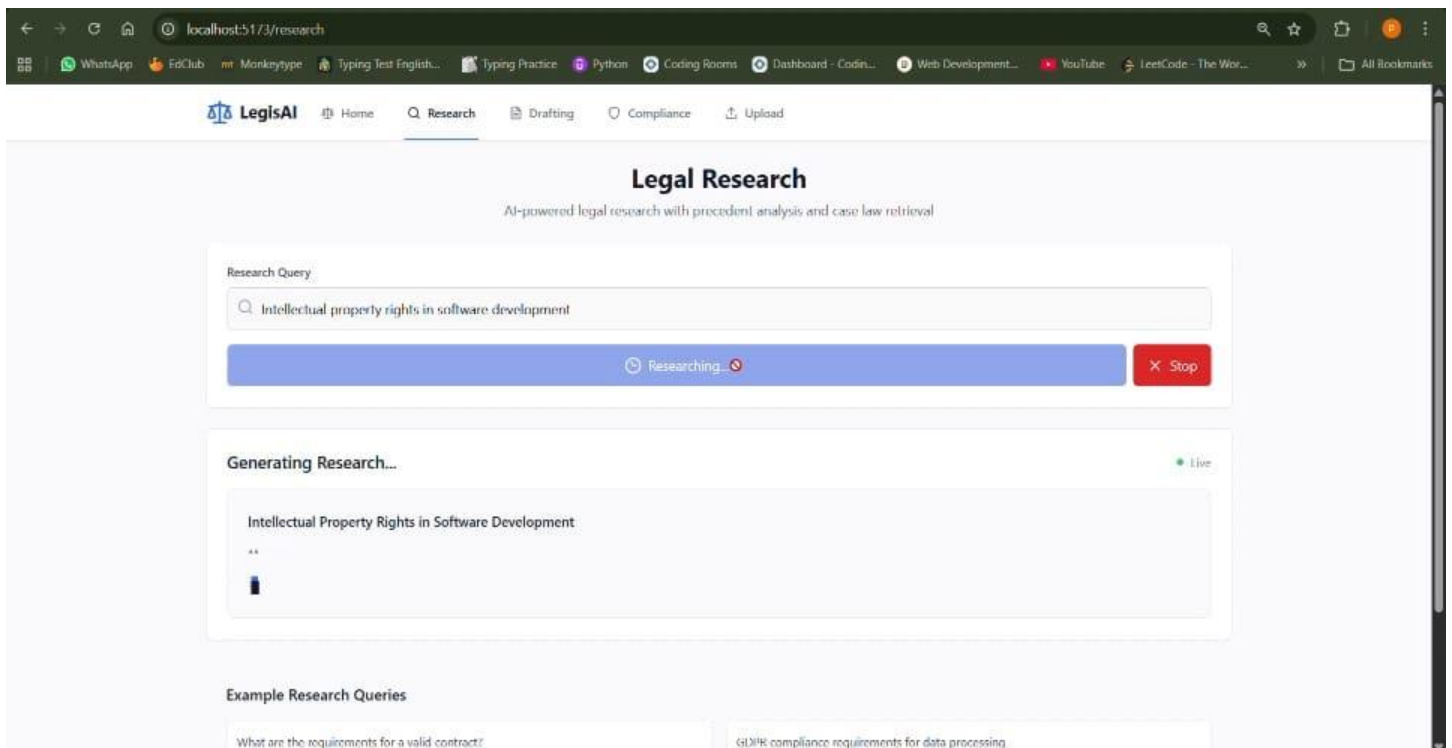
	implementation, still on progress. And working on FIASS.			
Sumith Padma	Developed an OCR Document Agent for extracting key legal entities and a Timeline Builder Agent that generates ordered case chronologies, with both components fully implemented and integrated into the legal intelligence workflow.	40	100	In Progress

Results Received so far:



Screenshots of Working Demo:





Code Complexity Metrics:

1. Overall Summary

- Total Functions: approx 100+
- Code Quality: Mostly A to C
- Main Hotspot: main.py

2. Critical Complexity (Rank F)

research_legal_query — 101

load_llm_models — 93

process_cuad_data — 53

3. Very High Complexity (Rank E)

train — 46

compute_predictions_logits — 44

4. Medium Complexity (Rank D)

generate_ai_response — 30
analyze_document — 29
parse_contrastive_summary — 30
parse_ratio_decidendi — 26
extract_clauses — 25
align_clauses — 26
detect_risk_changes — 24
search — 28
search_cases — 26

5. Rank Distribution

F: 3 | E: 2 | D: 9 | C: ~20 | B: ~30 | A: ~40

6. Module Snapshot

main.py — High
Services — Good
Scripts — Moderate
CUAD ML — High expected

7. Priority Refactoring

- research_legal_query
- load_llm_models
- generate_ai_response
- analyze_document

Pending Functionalities/Features/Research for Documentation (List or Screenshot):

- **Backend:** Retrieval and generation running locally.
- **Frontend:** Development in progress, working on implementing more features into the UI
- **Model integration:** Ollama and Ollama 3

Next Steps:

- Knowledge graph builder implementation
- Human-in-the-loop implementation
- Reasoning and self consistency implementation
- OCR and timeline builder implementation
- All the agents need to be fine tuned